Management Science Dissertation

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Part 2

INTEL

How much profit increase could Intel expect to achieve by adopting a more agile manufacturing process?

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TABLE OF CONTENTS

Executive summary *2*

Introduction *3*

Summary Part1 ..... **3**

Intel Current state **4**

Manufacturing Agility *5*

Defining Agility **5**

Operational agility **5**

R&D agility **5**

Measuring Agility *7*

For operation **8**

Resource utilization: RoNA 8

Asset utilization 8

Inventory turnover 8

For Research and development **9**

R&D costs over total revenue 9

Results and interpretation *10*

Strategy *12*

Recommendation **12**

Benefits of adopting business agility **14**

Feasibility study **15**

Risks involved. 15

Appendix *17*

Bibliography *21*

Sources *22*

EXECUTIVE SUMMARY

This paper recommends that Intel should draw its business model under operational and R&D agility criteria to increase its profit. By improving its manufacturing agility, it can potentially increase its net income by 30%.

In this paper we will measure the correlation and impact of the concerning parameters of operations and research and development to identify the main drivers of net income performance and measure the range and probability of their impact.

We will then study the industry risks and external factors that may adversely affect our results in a real-life scenario and how predictable and likely these are.

INTRODUCTION

SUMMARY PART1

In part 1 we identified Intel's reason of its inefficiency compared to its competitors. We measured the impact on open and closed innovation on Intel business model and how this can affect its revenue growth. We compared its specific revenue in from semiconductor sales with other companies with different business models: IDMs, fabless companies, and foundries. We compared them using scaled comparable companies to identify the weaknesses of Intel's business model.

Foundries like TSMC undergo less costs than IDMs thanks to their focus on manufacturing only. They can price for all tests and steps they need to make for the chip (part1). They have different customers, spreading the losses from a particular chip manufacture. They can more easily realize technological investment spreading costs across several fabless clients.

Fabless companies as AMD focus their investments on design and innovation, leading to high innovation rate and flexibility, not being bounded to a single fabrication house. By outsourcing their production they can leverage on manufacturing firms, to obtain high quality outputs in a timely manner allowing them to adapt to market changes and trends.

Around 2018, we have seen several IDMs moving form IDMs to fabless as AMD or modifying their business model as Samsung, who made the decision to divide its business in two distinct companies a foundry (Samsung electronics) and a product company (still Samsung).

Intel misses this change and is now thinking of adapting its new business model the IDM2.0, to accelerate its operational and R&D efficiency.

In part one we identified the inaptitude of Intel to adapt to changes, new constraints, and trends in the industry as a lack of business agility. The factors that drive the company's profitability are its operational and R&D agility.

In part 2, we will define, measure, and compare those parameters to determine the potential profit increase Intel can achieve by adopting a more agile manufacturing process.

We will first update the reader on Intel's current state and the recent development in the semiconductor industry as these drive our reasoning on the importance of implementing changes to Intel business model for its survival overtime.

INTEL CURRENT STATE

Intel disclosed a $7 billion operating losses for its manufacturing unit in 2023. The year before it was experiencing a $5.2 billion loss already. Its revenue for 2023 of $18.9 billion is down 31% from the year before.[1]

The Chief Executive Pat Gelsinger expect 2024 to be the year of operating losses for the company's chipmaking business. Those losses are attributed to bad decision making as the choice to not buy the new extreme ultraviolet (EUV) machines in 2022 that would have allowed them to create the new chips required by their consumers to use cryptocurrency and AI. Those machine although costing more than $150 million are also more cost effective than the earlier chip making tools used by Intel. (Intel discloses $7 billion operating loss for chip-making unit, 2024)

Those challenges result in a 7% stock drop in 2023 and an anticipated market value loss of $12 billion for 2024.

Intel difficulties when producing compelling bundle of chips and software to build AI applications lead Nvidia to control more than 83% market share in data centre chip during the year 2023. The remaining 17% share is held by Google's custom tensor processing units (TPUs), one of the few viable alternatives to the advanced AI chips made by Nvidia. (The Economic Times, n.d.)

"Our customers, first and foremost are asking for choice in the industry" said Intel vice president, strategy, and product management Jeni Barovian. "They are coming to us, and they are expecting that Intel, as a computing leader, will follow the wave of generative AI and deliver solutions that meet their needs. And they are looking for an open approach." (Intel reveals details of new AI chip to fight Nvidia dominance, 2024)

Credit: Intel 10-K 2023 annual report.

[1] See Appendix 1: Intel revenue comparison with other firms over the years.

MANUFACTURING AGILITY

DEFINING AGILITY

Agility is an iterative product-development process that promotes frequent feedback and the ability to switch focus and priority quickly to respond to the external environment requirements. Business agility applies the principle of agile development to the organization. This theory encourages companies to be more responsive to change, hasten the time to market, and reduce costs without sacrificing quality. (www.productplan.com, n.d.)

Agile development is extremely applicable to the semiconductor manufacturing industry where adapting to changes and following trends to satisfy customers is fundamental for a semiconductor factory and or designer company. The industry is rapidly evolving as technologies develop and customer requirements are high. Therefore, it is essential to adopt short-term development cycles and reduced overheads to accelerate iteration for product design and manufacture for those companies to more frequently improve and make successful decisions.

Those agile organization react in a timely manner, take advantage of opportunities, while responding adequately to threats in the marketplace. This allows them to remain customer centric under all circumstances. Seizing upon short-term opportunities, gives them the ability to become a first mover under new developments and to attract large market shares.

Nvidia and TSMC respectively adapted their designing and manufacturing processes in an agile manner to be effective at delivering the new chips used for AI applications, allowing them to acquire a major part of the market share.

In Q4 2023, Nvidia is dominant at 80% for the AIB GPU market report while Intel represents only 1%. TSMC for the same period maintained its dominant position in the foundry sector, commanding a 61% market share driven by smartphone restocking and strong demand from AI. (Counterpoint, n.d.)

Earlier we identified Intel's overall reason of loss of market share and decreased revenue as its inability to adapt to the new trend of integrating AI in its newly designed chips in this way we will concentrate our analysis on the operation and research and development agility. These appearing as the most important drivers of success for product design and manufacturing by semiconductor companies.

OPERATIONAL AGILITY

Operational agility measures the adaptability of a company to changes in demand, supply chain disruptions, and other unforeseen events in its everyday processes, technology, and operational procedures. This can be achieved by the adoption of effective new machineries, resources, and an agile workforce to achieve greater flexibility, and operating with high standards of accountability, expertise and transparency that attract customers.

Essentially in the semiconductor industry, manufacturers must adjust to the evolving requirements and deliver products to meet client needs. This process enables manufacturing companies important coast savings.

R&D AGILITY

Research and Development agility is the process whereby a company is effective at predicting new trends to deliver its product accordingly to the market demand. Semiconductor designer companies must

present high innovation rates after identifying new trends to design in a timely manner the product that will respond to consumer needs.

R&D agility is preponderant of a semiconductor company's success as the manufacturing process will require several iterations, test and design adjustments before the final product can be commercialized. Being rapid and adaptive in its innovation process will give the designer a leveraging advantage with manufacturers. Semiconductor foundries are appealed to integrate new advancements in their process and the high innovation rate of designers allows time to finalize the chips and deliver them in a timely manner on the market compared to competitors. By employing this methodology designers achieve important revenue growths during fast paced developments in the semiconductor industry.

We will now look at how to measure those factors of success for designer and manufacturer in the semiconductor industry.

MEASURING AGILITY

In this report we show the influence of each measure of Operation and R&D agility on the profit of a semiconductor company.

We perform a multiple linear regression model to measure the influence of each chosen measure of agility either operational or research and development on semiconductor companies' profit. This correlation test enables us to identify by how much Intel can increase its profit by improving its operational and R&D processes.

We run this test over a period of eight years and collected the data of ten companies from their 10-K annual report found on their investor relations website.

Data come from three IDMs, three foundries and four fabless companies. Those chosen are the most influential in the industry. Intel is an IDM company that means it is designing as well as manufacturing chops therefore the choice equally mix the type of companies between their characteristics will give a more accurate and comprehensive understanding of the overall impact of each measure on Intel's profit as a manufacturer and designer.

- IDMs:
    - Samsung (005390.KS)
    - Intel Corp. (INTC)
    - Texas Instruments
- Pure play foundries:
    - Taiwan Semiconductor Manufacturing Co. Ltd. (TSMC)
    - United Microelectronics (UMC) (initially an IDM)
    - GlobalFoundries.
- Fabless companies:
    - NVIDIA Corp. (NVDA)
    - Advanced Micro Devices (AMD)
    - Broadcom Inc. (AVGO)
    - Qualcomm Inc. (QCOM)

The test is concentrating on measuring operational and Research and Development agility for each of the previous companies. To achieve this, we identified four independent variables:
For operational agility:

- Resource utilisation: RoNA
- Asset Utilization
- Inventory Turnover

For R&D agility:

- R&D costs over the total revenue: investments in R&D.

Those independent variables are tested against the Net Income as the dependent variable for each company. We now expand further on the choice of each of these explanatory variables supposed to impact the profit.

FOR OPERATION

The following three measures are indicator of a company efficient resource utilization, reduced downtime, minimized retooling costs and optimized inventory management. All of those if well managed by a semiconductor company allow significant cost savings. Which lead to increased profitability over the longer term.

RESOURCE UTILIZATION: RONA

The RoNA (Return on Net Asset) ratio measure the resource utilisation rate of the company. It is highly relevant for semiconductor manufacturers as it measures the performance of a company to generate profit from the utilisation of its net asset. The objective being to generate high profit with minimal investment. It is a measure of financial performance.

$$RoNA = \frac{Net\ Income}{Fixed\ Assets + Working\ Capital}$$

With,

$$Working\ Capital = Current\ Assets - Current\ Liabilities$$

ASSET UTILIZATION

Asset Utilization is an efficiency metric used to represent the company's ability to generate revenue from its total assets. Relevant measure for semiconductor companies that rely heavily on expensive machinery and equipment to create their products. Optimizing this metric implies a high asset utilization rate revealing efficient production and response to market demands.

$$Asset\ Utilization = \frac{Revenue}{Total\ Assets}$$

INVENTORY TURNOVER

Inventory turnover calculates the number of times a company sold and replaced its inventory over a given period. The semiconductor industry's product life cycles are short and accelerated by technological advancements. Hence maintaining inventory levels low while meeting customer demands is essential to adapt quickly and with limiting costs to new trends and requirements. High inventory turnover ratio shows a reduced risk of inventory obsolescence. This allows the company to free up capital for investment in new technologies for future growth. Therefore, a highly relevant indicator of operational agility.

$$Inventory\ Turnover = \frac{Cost\ of\ Sales}{Average\ inventory}$$

For Research and development

The following metric will enable to attest of the product innovation rate, time to market and the ability of the company to meet evolving customer demands. It is a necessary indicator of a company's sustainable growth and competitiveness in the market.

R&D costs over total revenue

This ratio indicates how effectively a company allocates its resources and investments towards R&D activities to its total revenue. Calculating the proportion of revenue dedicated to R&D indicates the level of commitment of the company towards innovation which is a vital factor in the semiconductor industry to stay ahead of the new trends. Important investments in R&D also reflect the company's belief in its long-term growth prospects. A low ratio would indicate the well management of R&D investments that stay at low rates while generating high total revenues.

$$\frac{Research\ and\ Development\ Costs}{Total\ Revenue}$$

RESULTS AND INTERPRETATION

The previous parameters provide a comprehensive view of a semiconductor company's efficiency in terms of operational and research and development agility. The ability of the company to manage its resources, assets, and inventory as well as to innovate consistently and rapidly to meet new customer demands over the long term. Those are the most compelling drivers of sustainability and growth for semiconductor companies over the longer term.

After collection of all our original data[2], we calculate the mean of each variables using the AVERAGE function in excel and then standardized them with the excel STDEV.P function. This process allowed us to get homogeneity in our data. Certain companies being much larger than others were presenting extremely large number and would have too great an impact on our result.

The STDEV.P function calculates the standard deviation between the data based on the entire population. This measures the dispersion of the values compared to their average mean. Therefore, using these variations suppress the scaling issue that could influence our final comparisons.

We used our standardized data[3] to run the multiple linear regression in RStudio.

Using RStudio, a programming language for statistical computing and analysis we performed a multiple linear regression model.

Some of the standardized data appeared negative and the regression was at first skewed to the right. The data needed to be logged and we added 2 to the formula of the net income to make them positive and relevant in our analysis.

```r
# Fit the multiple linear regression model with log(zNI+2) on the dependent variable to remove skewness.
logzNImultiple.reg = lm(formula = log(zNI+2) ~ zRona + zAU + zRDCoverTtRev + zInvTurn,data = zvalues_operation_RD)
summary(logzNImultiple.reg)

summary(logzNImultiple.reg)$df

# Get the model residuals with the log on NI.
logzNImodel_residuals = logzNImultiple.reg$residuals
# Plot the result
hist(logzNImodel_residuals)

# Plot the residuals with the log on NI.
qqnorm(logzNImodel_residuals)
# Plot the Q-Q line with the log on NI.
qqline(logzNImodel_residuals)
```

[2] See Source Operational R&D original.
[3] See Source zvalues.operation.RD.

```
> library(readxl)
> zvalues_operation_RD <- read_excel("zvalues.operation.RD.xlsx",
+     col_types = c("numeric", "numeric", "numeric",
+         "numeric", "numeric"))
> View(zvalues_operation_RD)
> #### Correction:
>
> # Fit the multiple linear regression model with log(zNI+2) on the dependent variable to remove skewness.
> logzNImultiple.reg = lm(formula = log(zNI+2) ~ zRona + zAU + zRDCoverTtRev + zInvTurn,data = zvalues_operation_RD)
> summary(logzNImultiple.reg)

Call:
lm(formula = log(zNI + 2) ~ zRona + zAU + zRDCoverTtRev + zInvTurn,
    data = zvalues_operation_RD)

Residuals:
     Min       1Q   Median       3Q      Max
-0.54531 -0.27179 -0.02229  0.19694  0.91005

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)     0.58937    0.04287  13.747  < 2e-16 ***
zRona           0.15291    0.04448   3.438 0.000972 ***
zAU            -0.04797    0.04584  -1.047 0.298754
zRDCoverTtRev -0.14977    0.04353  -3.440 0.000965 ***
zInvTurn       -0.11826    0.04549  -2.600 0.011282 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.3787 on 73 degrees of freedom
Multiple R-squared:  0.3065,    Adjusted R-squared:  0.2685
F-statistic: 8.067 on 4 and 73 DF,  p-value: 1.921e-05

>
> summary(logzNImultiple.reg)$df
[1]  5 73  5
```

Picture of the code self-written in RStudio with the corresponding results

From these results we find out the most significant driver of profit increase is the R&D costs over total revenue with the lowest p-value. Its coefficient is negative meaning that as the ratio decreases profit increases. This is in line with our expectation as keeping relatively low R&D investments compared to total revenue is an indicator of well managed investment and profitable company over the longer term.

The second most significant variable is the RoNA ratio, positively correlated it shows that a high return on net asset increases net income. Which indicates the effective use of fixed assets as property, plants, and equipment as well as the available capital of the company used to finance its day-to-day operations.

The third important variable is the inventory turnover ratio, negatively correlated it shows that a longer time to sell its inventory for the company will lead to higher net income. Semiconductor companies usually have lower cost of goods sold as the major party of their cost are attributed to the process to create their products as property, plants, and equipment with heavy machineries, extremely costly reaching $150 million for example with the new EUV machines in 2022. They also have high inventory holding costs which shows that their product is relevant and effective over the longer term. Therefore, those negative correlation of the inventory turnover with the net income shows that a semiconductor companies with low cost of sales and high inventory holding cost are the one creating the most reliable and sustainable products that stay relevant to use for several years.

The overall multiple R-squares results imply that the independent variables represent more than 30% of the variation in my Net income response.

Therefore, Intel by adopting more agile operational and R&D processes would increase its profit by 30% over the next 10 years.

STRATEGY

RECOMMENDATION

The VUCA (Volatility, uncertainty, Complexity and Ambiguity) age shows how our world is tremendously changing where high capacity for transformation and innovation is necessary. Where the huge importance of redefining the methodology of Intel to integrate agile processes allowing high flexibility and resilience towards change and in the face of uncertainty and unpredictable events. (Burchardt and Maisch, 2019).

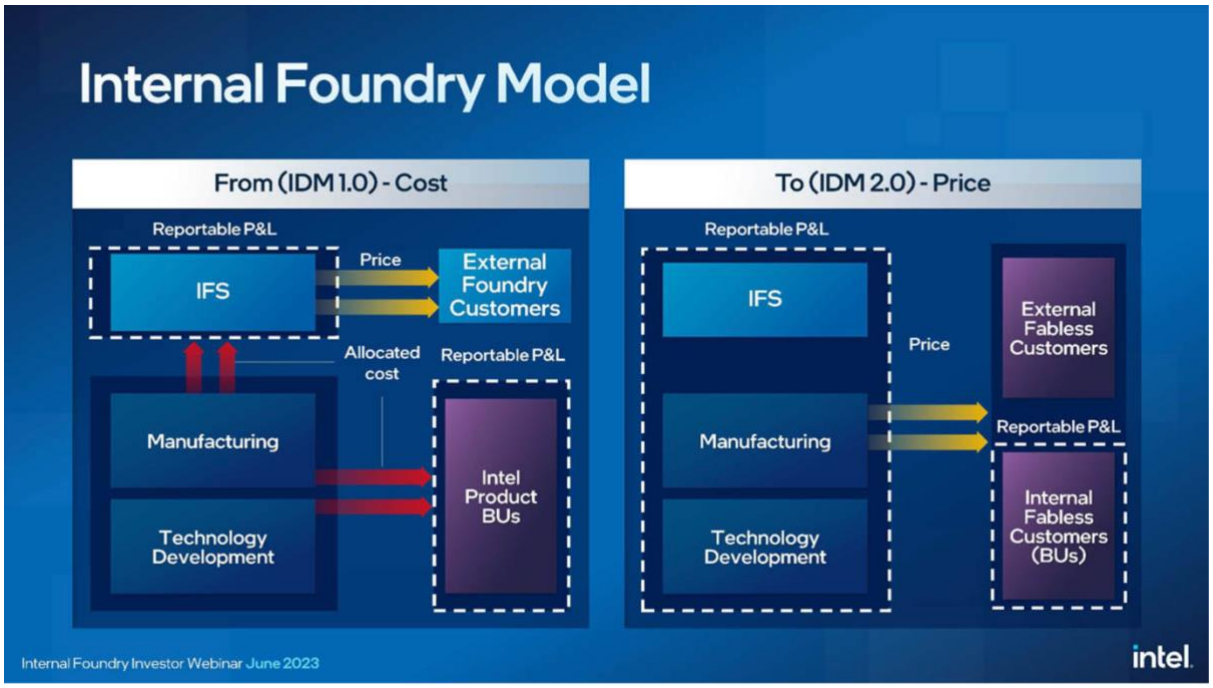
Credit: Intel (Investor relations webinar)

Semiconductor manufacturers process wafers in batches known as lots where the same product needs many iterations before giving birth to the final commercialized wafers. When an external company ask Intel to manufacture its designed chip a predefined number of iterations is priced by Intel to the designer. The process usually takes several months and if more iteration is needed than stated beforehand Intel as a manufacturer will have more costs to create the chip. If a product is extremely innovative it will be more complicated to manufacture, driving the number of iterations needed higher and the costs for intel too leading its margins to decrease. Intel calls those leading-edge wafers "hot lot" where it accepts to create them at no additional cost and in a few weeks for the client. Those hot lots represent important cost and disruptions for the manufacturing unit in terms of utilization and efficiency.

Intel would gain flexibility in its pricing process by separating its product and foundry business in two distinct entities and balance sheets. The designer would pay the manufacturer for each of the process steps and requirements. This would allow the firm more operational agility and viability of its manufacturing unit with a clearer vision of the cost incurred in each process.

After being manufactured chips are tested and sorted for failures. Jason Grebe, VP and GM of Intel Corporate Planning Group said *"We've increasingly grown our test times compared to our industry peers. Currently, we estimate our test times are 2 to 3x those of our competitors. Effectively, our lower-cost testing platform was subsidizing the growth in our test times."* However, those test time increase reveal themselves to be 3 times longer than their AMD competitors with no differentiation in the final error rate over similar product. Intel would therefore need to implement a more accurate test methodology, effective after shorter test time period. This would allow the company to reduce its time to market new product as

well as the draining costs of keeping those failed chips longer as inventory. Moreover, shorter more reliable tests would allow the company to quick reaction and design adaptation. Thus, achieving better operational and R&D agility as redesigning faster actual chips would clear the path to the innovation and manufacture of new chip designs and take the lead of new trends in the industry.



Therefore, Intel can save on operational and R&D cost. On one hand by effectively implementing its internal foundry model with two separate entities: manufacturing and product. And on the other hand, by reducing its time to market; decreasing test time while keeping the same accuracy level in its final product output failure rate.

Adopting an open innovation methodology sourcing external knowledge form other companies by effectively manufacturing for different customers would also allow intel to have a step in the new innovative trends of the chip industry. This will only be possible after improvement of their operation and R&D methodology.

BENEFITS OF ADOPTING BUSINESS AGILITY

Adopting Business agility will allow Intel to react in a timely manner. With efficient and effective responses to opportunities and threats while continuing satisfying its customers. This gives companies an important competitive advantage to stay ahead of the curve over the long term thanks to their resilience over time, in face of business cycles, economic events and extraordinary circumstances as the covid 19 pandemic in 2020.

They can react proactively and take advantage of those situations without incurring irremediable losses, reacting easily and actively to their setbacks. This constant actionable approach with high innovative rates gives them the potential to stay ahead of the competition and drive research over the long period with stable and effective products that satisfy their customers.

Cross functional collaboration is sometimes stated with business agility as this a way to share ideas and learn from other sectional entities of the company. This approach can be effective in the phase of research, idea generation, and the analysis of special trends. It gives a more comprehensive approach to unexpected circumstances while using the knowledge of diversified point of views. This process can also lead to decentralisation which comes with drawbacks if wrongly manage of dispersing people, reducing effectiveness. Sometimes suppressing the unified incentive between close employees that drive some companies towards their successful goal.

FEASIBILITY STUDY

RISKS INVOLVED.

The previous statements and recommendations come with risks and uncertainties. Hence the actual results may differ materially from those expressed, implied, or expected.

The semiconductor industry suffers from a high level of competition and rapid technological change that may affect the overall profitability of companies in the sector. The choice of investing in Research and Development and foundries equipment and facilities may not realize the favourable results predicted.

Implementing and developing semiconductors is extremely complex and there is uncertainty whether Intel would be able to achieve viable solutions for its customers.

The ability of Intel to time and scale their capital investment and secure with success favourable alternative is uncertain. For example, with financing arrangements as government grants. This depends heavily on the external bodies that would subsidize the company's activity.

Intel may not be able to implement new business strategies and invest appropriately into its different manufacturing and R&D facilities.

The company may wrongly analyse new trends the expected demand of customers leading to the creation of technologies that do not answer the market demand.

There is the risk of a change in demand for Intel products due to unexpected conflicts, as macroeconomic conditions and geopolitical tensions and conflicts, including geopolitical and trade tensions between the US and China, the impacts of Russia's war on Ukraine, tensions and conflict affecting Israel, and rising tensions between the US and Taiwan. (Intel 10-K Annual report, 2023)

Intel may not be able to catch the boat of the fast pace evolving trend that drive today's market with the need to create AI products with always more diverse capabilities.

Shortage, trade tensions and product defects or issues as Intel develop the next-generation products can negatively impact the manufacturing process leading to late deliveries. All of these disrupting the company's complex global supply chain.

Product defects, errata, and other supply issues may arise, as Intel develop its next-generation chips and implement new manufacturing process technologies. These could lead the company's customer to lose trust in the ability of Intel to deliver reliable products. Leading to poor publicity towards the company that would decrease demand for its future products. This could reduce its market share and will hit its profitability.

There are also potential security vulnerabilities in Intel products concerning the spread of cybersecurity threats and privacy risks. These could engage the company in costly regulatory proceedings especially regarding IP risks, extremely relevant to semiconductors' design methodology.

Intel also needs to attract, retain, and motivate key talent in order to make its business go forward in the long term. Stem students are currently highly demanded by the industry.

Catastrophic events also can be a risk for the potential success of the company if it touches a manufacturing entity for example.

The new ESG regulations are draining on costs and if not seriously considered can in the future lead the company to fail to adapt to those, leading to irremediable cost that could result in scandals and ultimately the company to go bankrupt.

Costs

All those investment decisions to make Intel profits increase depends on its accurate investment actions in its operations and Research and Development entities. The right management of the financial activities of the company, through the accounting of its investing and financing activities is compulsory for the company to gather a comprehensive and real view of its business state.

Agility for the company in terms of costs will come with investments to achieve acute and accurate accounting of the company's whole operation activities. Having clear knowledge at every point in time of the company's needs is essential for Intel to undertake the right action plan. Enabling the organization to react quickly, be resilient, innovative, profitable, and sustainable over time. All of these while satisfying, gaining, and retaining customers.

Hence those will increase demand for its product and restore its market share. Generating more profit and freeing up cash (in the form of free cash flow) for the company to reinvest in its Operations and Research and Development activities.

APPENDIX

From 2010 to 2022 intel is almost flat meanwhile Texas instrument is increasing.

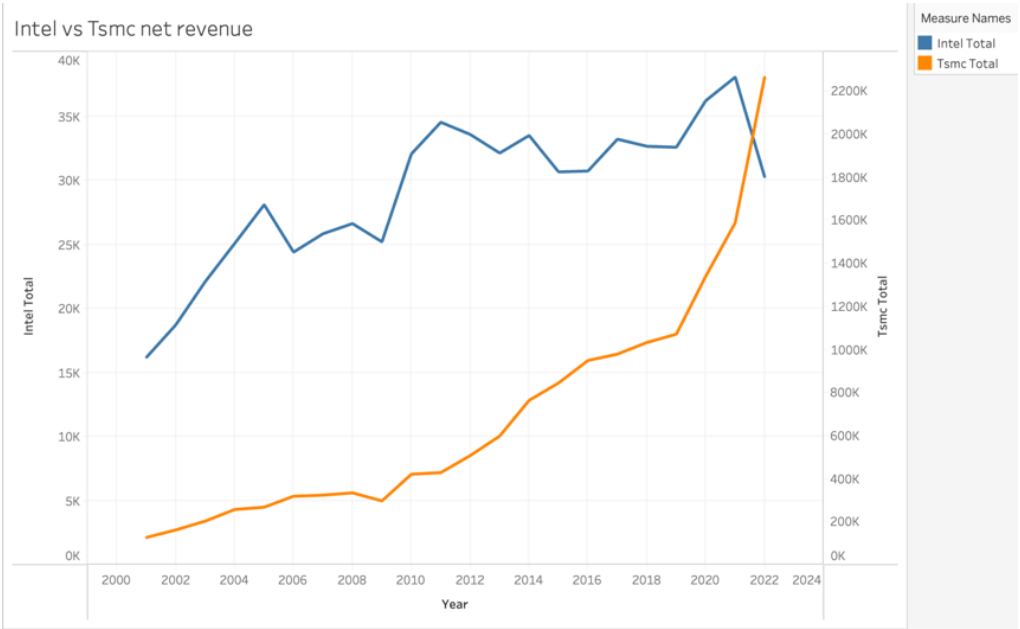
Not an industry trend as Intel is the only one who fails that much recently. All slow down during the pandemic and then go up but then intel goes a lot down. Fabless growth (and manufacturer) much stable and sustainable (seems less volatile) than IDM growth. Is it the association of manufacturing and designing that creates risk and volatility? is it the reason why intel wants to create IDM2.0 where its company is clearly divided in two parts:

- Manufacturing: includes Technology Developments, Manufacturing, and the Internal Foundry Service (IFS) to serve Intel products (the business unit) and external fabless customers.
- The Business Unit serves the final customers with Intel's ended products as computers software, semiconductors.

Under IDM 2.0, we now have two distinct balance sheets and Income statements one for Intel's Manufacturing unit ant one for its Business Unit. With the Manufacture setting prices to the BU and the fabless external customers.

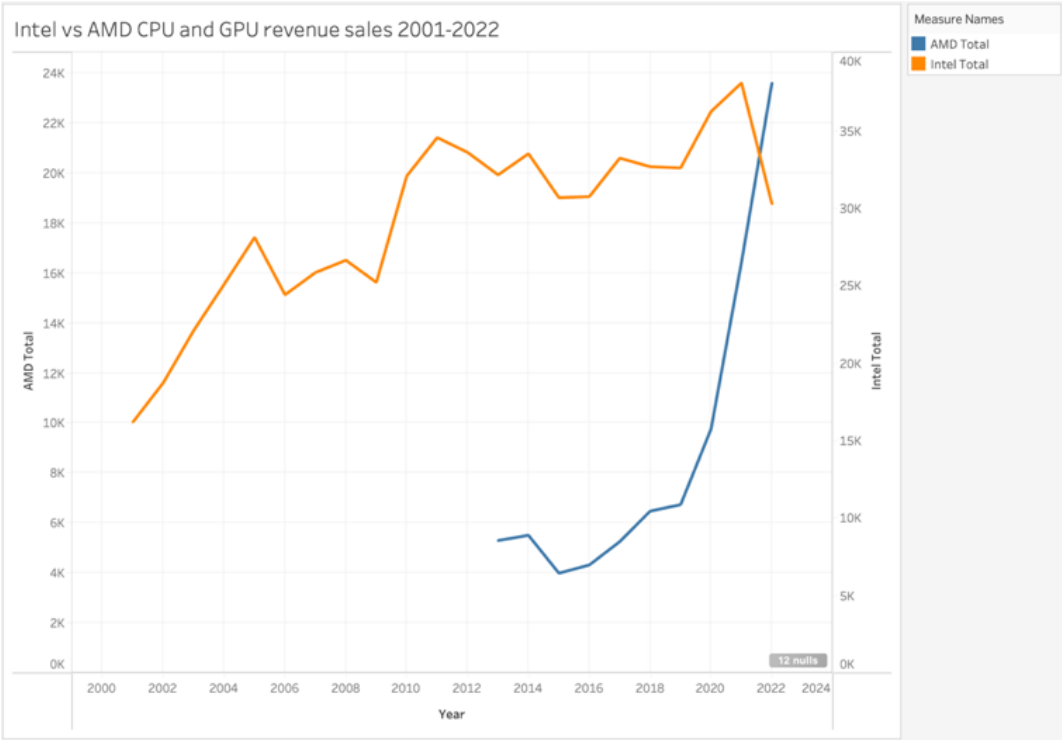
Under IDM 1.0 there was no distinction between the accounting of the manufacturing and the business unit.

Manufacturer:



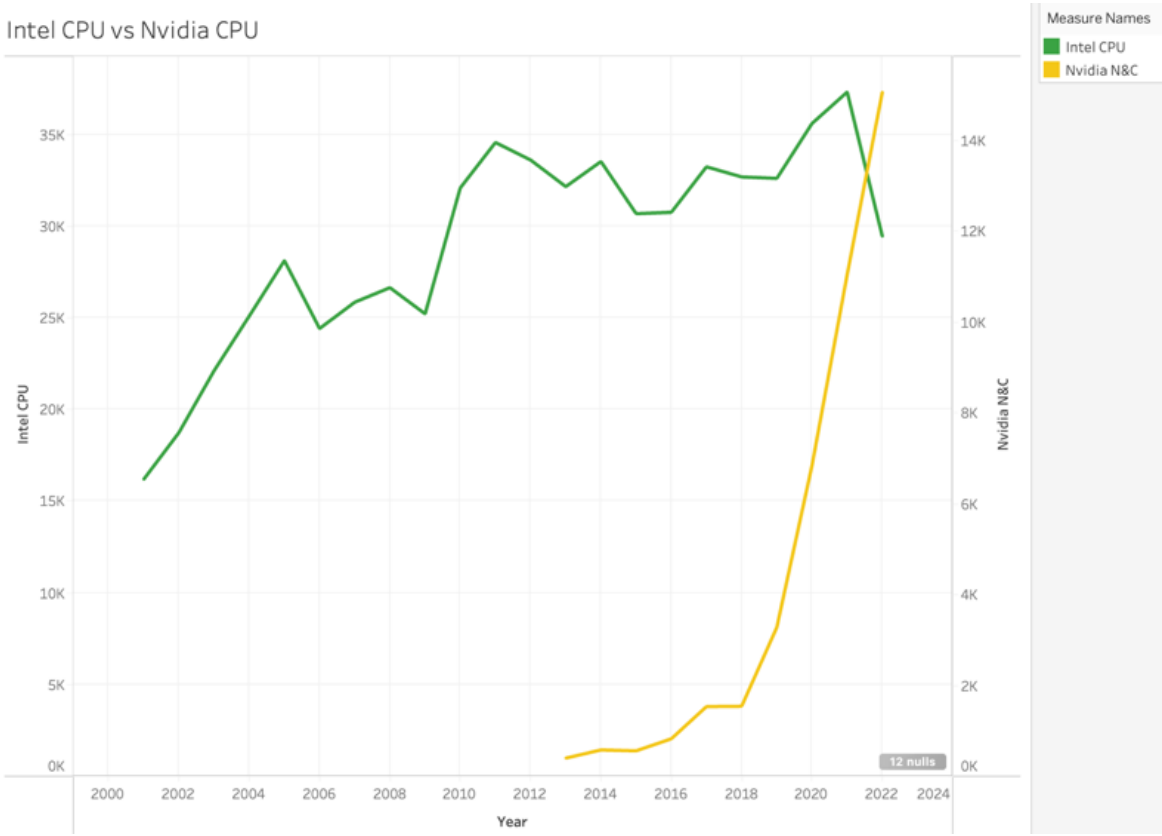
Credit: done myself using Tableau and Intel and TSMC annual 10-K reports.

Fabless:



Credit: done myself using Tableau and Intel and AMD annual 10-K reports.

Intel vs Nvidia revenue sales



Credit: done myself using Tableau and Intel and Nvidia annual 10-K reports.

IDMs:

Credit: done myself using Tableau and Intel and Samsung Electronics annual 10-K reports.

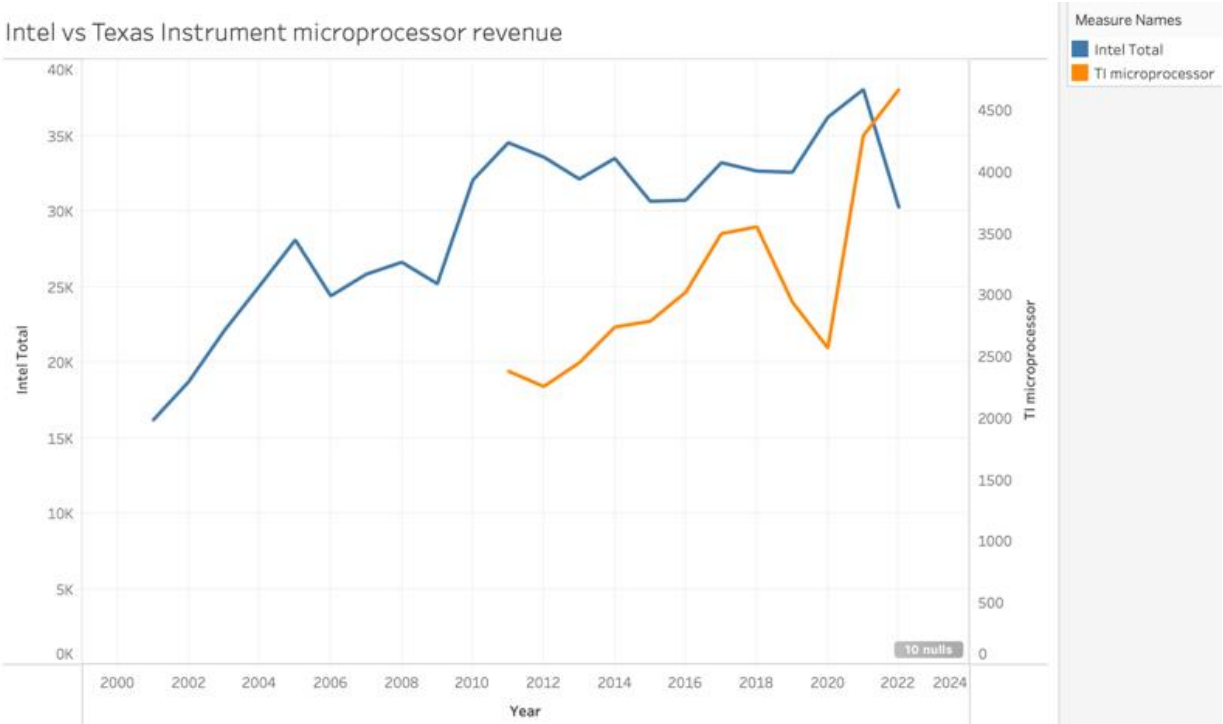
Samsung semiconductors:

In 2005, Samsung electronics was composed of the System LSI which was a large-scale integration with the association of the foundry and design units of the company.

In 2012, the foundry was detached to the System LSI.

In 2017, the Semiconductor Foundry, Memory vendor and OSAT were all independent entities within the company.

In the end, Samsung has 3 different companies: Semiconductor Foundry and Memory vendor and OSAT(outsourced semiconductor assembly and test). (Samsung global, n.d.)



Credit: done myself using Tableau and Intel and Samsung Electronics annual 10-K reports.

Texas instrument is almost fabless as it produces only 30% of its chips.(2017)

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SOURCES

Original data.

All collected from the 10-K annual reports found on the investor relations website for each company. Ratio are calculated using excel.

Company	Year	RONA	Net Income	Fixed Assets (PPE)	Working Capital	CA	CL	Asset Utilisatic	Revenue
Intel	2023	0.02	1,689,000,000.00	96,647,000,000.00	15,216,000,000.00	43,269,000,000.00	28,053,000,000.00	0.28	54,228,000,000.00
Intel	2022	0.08	8,014,000,000.00	80,860,000,000.00	18,252,000,000.00	50,407,000,000.00	32,155,000,000.00	0.35	63,054,000,000.00
Intel	2021	0.21	19,868,000,000.00	63,245,000,000.00	31,096,000,000.00	58,558,000,000.00	27,462,000,000.00	0.47	79,024,000,000.00
Intel	2020	0.26	20,899,000,000.00	56,584,000,000.00	22,495,000,000.00	47,249,000,000.00	24,754,000,000.00	0.51	77,867,000,000.00
Intel	2019	0.33	21,048,000,000.00	55,386,000,000.00	8,929,000,000.00	31,239,000,000.00	22,310,000,000.00	0.53	71,965,000,000.00
Intel	2018	0.34	21,053,000,000.00	48,976,000,000.00	12,161,000,000.00	28,787,000,000.00	16,626,000,000.00	0.55	70,848,000,000.00
Intel	2017	0.18	9,601,000,000.00	41,109,000,000.00	12,079,000,000.00	29,500,000,000.00	17,421,000,000.00	0.51	62,761,000,000.00
Intel	2016	0.20	10,316,000,000.00	36,171,000,000.00	15,206,000,000.00	35,508,000,000.00	20,302,000,000.00	0.52	59,387,000,000.00
SamsungEl.	2023	0.05	11,857,333,000.00	143,368,344,000.00	92,041,393,000.00	150,014,208,000.00	57,972,815,000.00	0.56	196,770,000,000.00
SamsungEl.	2022	0.18	42,610,233,000.00	128,659,991,000.00	107,283,964,000.00	167,266,852,000.00	59,982,888,000.00	0.66	233,130,000,000.00
SamsungEl.	2021	0.14	34,881,111,000.00	131,045,055,000.00	113,666,766,000.00	190,685,555,000.00	77,018,789,000.00	0.67	240,710,000,000.00
SamsungEl.	2020	0.10	23,081,768,000.00	112,711,289,000.00	107,168,357,000.00	173,250,348,000.00	66,081,991,000.00	0.58	203,430,000,000.00
SamsungEl.	2019	0.09	18,652,605,000.00	102,813,888,000.00	100,906,506,000.00	155,634,050,000.00	54,727,544,000.00	0.65	198,030,000,000.00
SamsungEl.	2018	0.20	38,049,231,000.00	99,031,047,000.00	90,621,655,000.00	149,895,684,000.00	59,274,029,000.00	0.73	221,670,000,000.00
SamsungEl.	2017	0.22	37,297,884,000.00	98,725,136,000.00	70,558,776,000.00	129,949,219,000.00	59,390,443,000.00	0.76	214,570,000,000.00
SamsungEl.	2016	0.10	19,594,260,000.00	78,867,346,000.00	121,939,593,000.00	141,429,704,000.00	47,165,446,000.00	0.80	174,047,940,000.00
TexasI.	2023	0.30	6,476,000,000.00	9,999,000,000.00	11,802,000,000.00	15,122,000,000.00	3,320,000,000.00	0.54	17,519,000,000.00
TexasI.	2022	0.49	8,709,000,000.00	6,876,000,000.00	11,036,000,000.00	14,021,000,000.00	2,985,000,000.00	0.74	20,028,000,000.00
TexasI.	2021	0.48	7,736,000,000.00	5,141,000,000.00	11,116,000,000.00	13,685,000,000.00	2,569,000,000.00	0.74	18,344,000,000.00
TexasI.	2020	0.50	5,568,000,000.00	3,269,000,000.00	7,849,000,000.00	10,239,000,000.00	2,390,000,000.00	0.75	14,461,000,000.00
TexasI.	2019	0.50	5,017,000,000.00	3,303,000,000.00	6,638,000,000.00	8,761,000,000.00	2,123,000,000.00	0.80	14,383,000,000.00
TexasI.	2018	0.63	5,580,000,000.00	3,183,000,000.00	5,623,000,000.00	8,097,000,000.00	2,474,000,000.00	0.92	15,784,000,000.00
TexasI.	2017	0.40	3,682,000,000.00	2,664,000,000.00	6,476,000,000.00	8,734,000,000.00	2,258,000,000.00	0.85	14,961,000,000.00
TexasI.	2016	0.47	3,595,000,000.00	2,512,000,000.00	5,193,000,000.00	7,457,000,000.00	2,264,000,000.00	0.81	13,370,000,000.00
TSMC	2023	0.20	26,879,000,000.00	98,992,200,000.00	35,153,200,000.00	65,806,300,000.00	30,653,100,000.00	0.40	69,298,000,000.00
TSMC	2022	0.26	32,311,200,000.00	87,661,500,000.00	34,700,000,000.00	66,804,300,000.00	32,104,300,000.00	0.45	73,670,400,000.00
TSMC	2021	0.21	21,354,000,000.00	71,201,100,000.00	30,595,500,000.00	57,933,400,000.00	27,337,900,000.00	0.42	57,224,800,000.00
TSMC	2020	0.25	18,188,900,000.00	55,398,500,000.00	16,392,000,000.00	38,895,500,000.00	22,503,500,000.00	0.49	47,769,370,000.00
TSMC	2019	0.22	11,833,800,000.00	45,214,900,000.00	7,497,500,000.00	27,503,000,000.00	20,005,500,000.00	0.48	35,773,600,000.00
TSMC	2018	0.22	11,860,600,000.00	35,022,900,000.00	19,433,000,000.00	31,090,500,000.00	11,657,500,000.00	0.49	33,693,700,000.00
TSMC	2017	0.23	11,639,600,000.00	35,848,300,000.00	15,867,500,000.00	28,920,500,000.00	13,053,000,000.00	0.49	32,977,100,000.00
TSMC	2016	0.23	10,238,100,000.00	30,795,600,000.00	14,489,000,000.00	25,238,600,000.00	10,749,600,000.00	0.50	29,256,500,000.00
UMC	2023	0.18	1,958,000,000.00	6,710,791,000.00	4,019,836,000.00	6,929,255,000.00	2,909,419,000.00	0.42	7,180,000,000.00
UMC	2022	0.29	2,911,774,000.00	5,564,011,000.00	4,548,074,000.00	8,212,530,000.00	3,664,456,000.00	0.53	9,069,485,000.00
UMC	2021	0.20	1,847,384,000.00	4,684,272,000.00	4,506,995,000.00	8,409,280,000.00	3,902,285,000.00	0.47	7,678,840,000.00
UMC	2020	0.11	814,129,000.00	4,728,442,000.00	3,020,825,000.00	5,851,350,000.00	2,830,525,000.00	0.48	6,297,041,000.00
UMC	2019	0.04	272,655,000.00	5,027,553,000.00	2,704,046,000.00	5,143,207,000.00	2,439,161,000.00	0.40	4,954,919,000.00
UMC	2018	0.03	250,824,000.00	5,646,736,000.00	2,978,653,000.00	4,612,657,000.00	1,634,004,000.00	0.42	4,941,280,000.00
UMC	2017	0.04	326,474,000.00	6,941,352,000.00	1,702,523,000.00	4,698,987,000.00	2,996,464,000.00	0.38	5,036,596,000.00
UMC	2016	0.03	266,085,000.00	6,943,932,000.00	1,162,960,000.00	3,409,556,000.00	2,246,596,000.00	0.38	4,563,892,000.00
GlobalF.	2023	0.08	1,020,000,000.00	10,596,000,000.00	2,441,000,000.00	5,800,000,000.00	3,359,000,000.00	0.41	7,392,000,000.00
GlobalF.	2022	0.11	1,448,000,000.00	10,596,000,000.00	2,441,000,000.00	5,800,000,000.00	3,359,000,000.00	0.45	8,108,000,000.00
GlobalF.	2021	-0.02	-250,000,000.00	8,713,000,000.00	2,128,000,000.00	5,291,000,000.00	3,163,000,000.00	0.44	6,585,000,000.00
GlobalF.	2020	-0.14	-1,347,571,000.00	8,226,202,000.00	1,090,654,000.00	2,986,724,000.00	1,896,070,000.00	0.39	4,850,505,000.00
GlobalF.	2019	-0.13	-1,371,186,000.00	9,611,269,000.00	1,178,083,000.00	3,514,150,000.00	2,336,067,000.00	0.40	5,812,788,000.00
GlobalF.	2018	-0.25	-2,701,603,000.00	9,611,269,000.00	1,178,083,000.00	3,514,150,000.00	2,336,067,000.00	0.43	6,196,039,000.00
Nvidia	2023	0.79	29,760,000,000.00	3,914,000,000.00	33,714,000,000.00	44,345,000,000.00	10,631,000,000.00	0.93	60,922,000,000.00
Nvidia	2022	0.21	4,368,000,000.00	3,807,000,000.00	16,510,000,000.00	23,073,000,000.00	6,563,000,000.00	0.65	26,974,000,000.00
Nvidia	2021	0.36	9,752,000,000.00	2,778,000,000.00	24,494,000,000.00	28,829,000,000.00	4,335,000,000.00	0.61	26,914,000,000.00
Nvidia	2020	0.30	4,332,000,000.00	2,149,000,000.00	12,130,000,000.00	16,055,000,000.00	3,925,000,000.00	0.58	16,675,000,000.00
Nvidia	2019	0.21	2,796,000,000.00	1,674,000,000.00	11,906,000,000.00	13,690,000,000.00	1,784,000,000.00	0.63	10,918,000,000.00
Nvidia	2018	0.39	4,141,000,000.00	1,404,000,000.00	9,228,000,000.00	10,557,000,000.00	1,329,000,000.00	0.88	11,716,000,000.00
Nvidia	2017	0.33	3,047,000,000.00	997,000,000.00	8,102,000,000.00	9,255,000,000.00	1,153,000,000.00	0.86	9,714,000,000.00
Nvidia	2016	0.23	1,666,000,000.00	521,000,000.00	6,748,000,000.00	8,536,000,000.00	1,788,000,000.00	0.70	6,910,000,000.00
Broadcom	2023	0.90	14,082,000,000.00	2,154,000,000.00	13,442,000,000.00	20,847,000,000.00	7,405,000,000.00	0.49	35,819,000,000.00
Broadcom	2022	0.82	11,223,000,000.00	2,223,000,000.00	11,452,000,000.00	18,504,000,000.00	7,052,000,000.00	0.45	33,203,000,000.00
Broadcom	2021	0.51	6,437,000,000.00	2,348,000,000.00	10,305,000,000.00	16,586,000,000.00	6,281,000,000.00	0.36	27,450,000,000.00
Broadcom	2020	0.33	2,663,000,000.00	2,509,000,000.00	5,524,000,000.00	11,895,000,000.00	6,371,000,000.00	0.31	23,888,000,000.00
Broadcom	2019	0.48	2,695,000,000.00	2,565,000,000.00	3,018,000,000.00	9,917,000,000.00	6,899,000,000.00	0.33	22,597,000,000.00
Broadcom	2018	1.30	12,259,000,000.00	2,635,000,000.00	6,769,000,000.00	9,107,000,000.00	2,338,000,000.00	0.42	20,848,000,000.00
Broadcom	2017	0.11	1,692,000,000.00	2,599,000,000.00	13,294,000,000.00	15,823,000,000.00	2,529,000,000.00	0.32	17,636,000,000.00
Broadcom	2016	-0.27	-1,739,000,000.00	2,509,000,000.00	4,047,000,000.00	7,125,000,000.00	3,078,000,000.00	0.26	13,240,000,000.00
Qualcomm	2023	0.40	7,232,000,000.00	5,042,000,000.00	12,836,000,000.00	22,464,000,000.00	9,628,000,000.00	0.70	35,820,000,000.00
Qualcomm	2022	0.92	12,936,000,000.00	5,168,000,000.00	8,858,000,000.00	20,724,000,000.00	11,866,000,000.00	0.90	44,200,000,000.00
Qualcomm	2021	0.71	9,043,000,000.00	4,559,000,000.00	8,124,000,000.00	20,075,000,000.00	11,951,000,000.00	0.81	33,566,000,000.00
Qualcomm	2020	0.38	5,198,000,000.00	3,711,000,000.00	9,847,000,000.00	18,519,000,000.00	8,672,000,000.00	0.66	23,531,000,000.00
Qualcomm	2019	0.40	4,386,000,000.00	3,081,000,000.00	7,830,000,000.00	16,765,000,000.00	8,935,000,000.00	0.74	24,273,000,000.00
Qualcomm	2018	-0.55	-4,964,000,000.00	2,975,000,000.00	5,995,000,000.00	17,384,000,000.00	11,389,000,000.00	0.69	22,611,000,000.00
Qualcomm	2017	0.07	2,445,000,000.00	3,216,000,000.00	32,686,000,000.00	43,593,000,000.00	10,907,000,000.00	0.34	22,258,000,000.00
Qualcomm	2016	0.32	5,705,000,000.00	2,306,000,000.00	15,670,000,000.00	22,981,000,000.00	7,311,000,000.00	0.45	23,554,000,000.00
AMD	2023	0.07	854,000,000.00	1,589,000,000.00	10,079,000,000.00	16,768,000,000.00	6,689,000,000.00	0.33	22,680,000,000.00
AMD	2022	0.13	1,320,000,000.00	1,513,000,000.00	8,650,000,000.00	15,019,000,000.00	6,369,000,000.00	0.35	23,601,000,000.00
AMD	2021	0.63	3,162,000,000.00	702,000,000.00	4,343,000,000.00	8,583,000,000.00	4,240,000,000.00	1.32	16,434,000,000.00
AMD	2020	0.57	2,490,000,000.00	641,000,000.00	3,726,000,000.00	6,143,000,000.00	2,417,000,000.00	1.09	9,763,000,000.00
AMD	2019	0.12	341,000,000.00	500,000,000.00	2,238,000,000.00	4,597,000,000.00	2,359,000,000.00	1.12	6,731,000,000.00
AMD	2018	0.18	337,000,000.00	348,000,000.00	1,556,000,000.00	3,540,000,000.00	1,984,000,000.00	1.42	6,475,000,000.00
AMD	2017	0.00	-33.00	261,000,000.00	1,121,000,000.00	2,634,000,000.00	1,513,000,000.00	1.48	5,253,000,000.00
AMD	2016	0.00	-498.00	164,000,000.00	1,184,000,000.00	2,530,000,000.00	1,346,000,000.00	1.30	4,319,000,000.00

TA	R&D costs/tot	COS	R&D costs	Inventory	Inventory turnover
191,572,000,000.00	0.30	32,517,000,000.00	16,046,000,000.00	11,127,000,000.00	
182,103,000,000.00	0.28	36,188,000,000.00	17,528,000,000.00	13,224,000,000.00	2.92
168,406,000,000.00	0.19	35,209,000,000.00	15,190,000,000.00	10,776,000,000.00	2.74
153,091,000,000.00	0.17	34,255,000,000.00	13,556,000,000.00	8,427,000,000.00	3.27
136,524,000,000.00	0.19	29,825,000,000.00	13,362,000,000.00	8,744,000,000.00	4.06
127,963,000,000.00	0.19	27,111,000,000.00	13,543,000,000.00	7,253,000,000.00	3.41
123,249,000,000.00	0.21	23,663,000,000.00	13,035,000,000.00	6,983,000,000.00	3.74
113,327,000,000.00	0.21	23,154,000,000.00	12,685,000,000.00	5,553,000,000.00	3.39
352,450,000,000.00	0.11	138,110,266,000.00	20,954,516,620.39	39,526,134,000.00	4.17
355,630,000,000.00	0.08	145,501,003,000.00	18,425,364,645.67	39,956,410,000.00	3.49
358,530,000,000.00	0.07	145,451,851,000.00	16,707,163,319.51	36,172,043,000.00	3.64
348,140,000,000.00	0.08	126,290,010,000.00	15,690,880,069.08	28,007,314,000.00	4.02
304,890,000,000.00	0.08	126,335,995,000.00	14,930,744,987.91	22,966,437,000.00	4.51
304,000,000,000.00	0.06	113,598,417,000.00	13,790,175,991.29	24,869,754,000.00	5.50
283,050,000,000.00	0.06	114,307,653,000.00	12,424,325,262.07	22,088,128,000.00	4.57
217,490,000,000.00	0.06	103,702,512,000.00	10,937,523,500.51	15,824,248,000.00	5.18
32,348,000,000.00	0.11	6,500,000,000.00	1,863,000,000.00	3,999,000,000.00	6.55
27,207,000,000.00	0.08	6,257,000,000.00	1,670,000,000.00	2,757,000,000.00	1.63
24,676,000,000.00	0.08	5,968,000,000.00	1,554,000,000.00	1,910,000,000.00	2.27
19,351,000,000.00	0.11	5,192,000,000.00	1,530,000,000.00	1,955,000,000.00	3.12
18,018,000,000.00	0.11	5,219,000,000.00	1,544,000,000.00	2,001,000,000.00	2.66
17,137,000,000.00	0.10	5,507,000,000.00	1,559,000,000.00	2,217,000,000.00	2.61
17,642,000,000.00	0.10	5,347,000,000.00	1,508,000,000.00	1,957,000,000.00	2.48
16,431,000,000.00	0.10	5,113,000,000.00	1,356,000,000.00	1,790,000,000.00	2.73
173,312,000,000.00	0.08	31,628,000,000.00	5,846,000,000.00	8,282,046,000.00	2.86
161,980,000,000.00	0.07	29,792,900,000.00	5,312,900,000.00	7,196,500,000.00	3.82
134,680,000,000.00	0.08	27,681,300,000.00	4,496,600,000.00	6,961,200,000.00	4.14
98,340,000,000.00	0.08	22,368,600,000.00	3,899,100,000.00	4,891,500,000.00	3.98
75,300,000,000.00	0.09	19,300,800,000.00	3,056,500,000.00	2,774,400,000.00	4.57
68,400,000,000.00	0.08	17,428,600,000.00	2,806,100,000.00	3,372,500,000.00	6.96
67,170,000,000.00	0.08	16,282,600,000.00	2,723,800,000.00	2,492,600,000.00	5.17
58,140,000,000.00	0.08	14,601,100,000.00	2,197,800,000.00	1,502,500,000.00	6.53
17,293,760,000.00	0.06	4,559,000,000.00	426,000,000.00	1,155,314,000.00	9.72
17,072,750,000.00	0.05	4,976,925,000.00	421,527,000.00	1,011,063,000.00	3.95
16,256,500,000.00	0.06	5,081,521,000.00	466,288,000.00	829,531,000.00	4.92
13,050,360,000.00	0.07	4,908,255,000.00	459,241,000.00	803,151,000.00	6.13
12,245,460,000.00	0.08	4,242,282,000.00	396,531,000.00	726,005,000.00	6.11
11,845,710,000.00	0.09	4,195,118,000.00	425,519,000.00	594,679,000.00	5.84
13,196,090,000.00	0.09	4,123,716,000.00	461,187,000.00	615,975,000.00	7.05
11,858,850,000.00	0.09	3,626,256,000.00	417,665,000.00	524,624,000.00	6.69
17,841,000,000.00	0.06	5,291,000,000.00	428,000,000.00	1,339,000,000.00	6.91
17,841,000,000.00	0.06	5,869,000,000.00	482,000,000.00	1,339,000,000.00	3.95
15,028,000,000.00	0.07	5,572,000,000.00	478,000,000.00	1,121,000,000.00	4.38
					4.97
12,321,630,000.00	0.10	5,563,225,000.00	475,769,000.00	919,519,000.00	6.05
14,497,510,000.00	0.10	6,345,032,000.00	582,974,000.00	351,923,000.00	18.03
14,497,510,000.00	0.15	6,646,090,000.00	926,220,000.00	351,923,000.00	18.89
65,728,000,000.00	0.14	16,621,000,000.00	8,675,000,000.00	5,282,000,000.00	3.15
41,182,000,000.00	0.27	11,618,000,000.00	7,339,000,000.00	5,159,000,000.00	2.25
44,187,000,000.00	0.20	9,439,000,000.00	5,268,000,000.00	2,605,000,000.00	3.62
28,791,000,000.00	0.24	6,279,000,000.00	3,924,000,000.00	1,826,000,000.00	3.44
17,315,000,000.00	0.26	4,150,000,000.00	2,829,000,000.00	979,000,000.00	4.24
13,292,000,000.00	0.20	4,545,000,000.00	2,376,000,000.00	1,575,000,000.00	2.89
11,241,000,000.00	0.18	3,892,000,000.00	1,797,000,000.00	796,000,000.00	4.89
9,841,000,000.00	0.21	2,847,000,000.00	1,463,000,000.00	794,000,000.00	3.59
72,861,000,000.00	0.15	11,129,000,000.00	5,253,000,000.00	1,898,000,000.00	5.86
73,249,000,000.00	0.15	11,108,000,000.00	4,919,000,000.00	1,925,000,000.00	5.77
75,570,000,000.00	0.18	10,606,000,000.00	4,854,000,000.00	1,297,000,000.00	8.18
75,933,000,000.00	0.21	10,372,000,000.00	4,968,000,000.00	1,003,000,000.00	10.34
67,493,000,000.00	0.21	10,114,000,000.00	4,696,000,000.00	874,000,000.00	11.57
50,124,000,000.00	0.18	10,115,000,000.00	3,768,000,000.00	1,124,000,000.00	9.00
54,418,000,000.00	0.19	9,127,000,000.00	3,302,000,000.00	1,447,000,000.00	6.31
49,966,000,000.00	0.20	7,300,000,000.00	2,674,000,000.00	1,400,000,000.00	5.21
51,040,000,000.00	0.25	15,869,000,000.00	8,818,000,000.00	6,422,000,000.00	2.47
49,014,000,000.00	0.19	18,635,000,000.00	8,194,000,000.00	6,341,000,000.00	2.94
41,240,000,000.00	0.21	14,262,000,000.00	7,176,000,000.00	3,228,000,000.00	4.42
35,594,000,000.00	0.25	9,255,000,000.00	5,975,000,000.00	2,598,000,000.00	3.56
32,957,000,000.00	0.22	8,599,000,000.00	5,398,000,000.00	1,400,000,000.00	6.14
32,718,000,000.00	0.25	10,244,000,000.00	5,625,000,000.00	1,693,000,000.00	6.05
65,486,000,000.00	0.25	9,792,000,000.00	5,485,000,000.00	2,035,000,000.00	4.81
52,359,000,000.00	0.22	9,749,000,000.00	5,151,000,000.00	1,556,000,000.00	6.27
67,885,000,000.00	0.26	12,220,000,000.00	5,872,000,000.00	4,351,000,000.00	2.81
67,580,000,000.00	0.21	12,998,000,000.00	5,005,000,000.00	3,771,000,000.00	3.45
12,419,000,000.00	0.17	8,505,000,000.00	2,845,000,000.00	1,955,000,000.00	4.35
8,962,000,000.00	0.20	5,416,000,000.00	1,983,000,000.00	1,399,000,000.00	3.87
6,028,000,000.00	0.23	3,863,000,000.00	1,547,000,000.00	982,000,000.00	3.93
4,556,000,000.00	0.22	4,028,000,000.00	1,434,000,000.00	845,000,000.00	4.77
3,552,000,000.00	0.23	3,266,000,000.00	1,196,000,000.00	694,000,000.00	4.71
3,321,000,000.00	0.23	3,316,000,000.00	1,008,000,000.00	751,000,000.00	4.42

Standardized data.

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